Hind Mukhtar

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Education

University of Ottawa, PhD in Electrical & Computer Engineering, AI-Enabled Wireless Communications — Sept 2021 – Sept 2026

University of Ottawa, Master in Electrical & Computer Engineering — Sept 2019 – May 2021

Queen's University, Bachelor in Electrical Engineering — Sept 2014 – May 2019

Licensed Professional Engineer (P.Eng), Ontario — Dec 2024 – Present

Technical Skills

Programming: Python, SQL, C/C++ (familiar)
ML Frameworks: PyTorch, Tensorflow
LLMs & GenAI: Google Gemini, Abacus AI, RAG, Agent Orchestration, Prompt Engineering
Data Visualization: Power BI, Tableau, Plotly, Dash, SQL Server Reporting Services (SSRS)
Cloud Services: AWS (EC2 GPU, Athena, Redshift, S3), Azure (Functions, DevOps)

Experience

Machine Learning Engineer, Gogo Business Aviation — Jan 2026 – Present

- Developed Text-to-SQL AI agents to enable natural language querying of enterprise datasets, empowering business users to self-serve analytics, reducing reporting development time.
- Developed a RAG-powered sales assistant, integrating Salesforce, AWS Athena, and internal documentation to provide customer insights, retrieve pricing information, generate sales quotes and answer customer-related questions.
- Developed cloud-hosted connectivity performance monitoring platforms for in-flight Wi-Fi connectivity, analyzing router telemetry, client device activity and network performance to improve user connectivity experience through real time diagnostics and anomaly detection.
- Applied large language models and multi-agent orchestration to automate fault triage and log analysis across high-volume operational network data, reducing manual investigation time during connectivity incident response.
- Designed, bench-marked and evaluated ML models on 10M+ records using AWS GPU clusters, implementing reproducible training pipelines and improving predictive performance across large-scale datasets.

Senior Data Scientist, Gogo Business Aviation — Jan 2023 – Jan 2026

- Led real-time monitoring architecture for LEO, LTE and 5G product launches, engineering spatial-temporal pipelines across satellite telemetry and geolocation data to power proactive fault detection.
- Built multi-variable forecasting models incorporating historical demand, projected growth, and allocation constraints to inform global satellite bandwidth allocation and provider contract strategy.
- Built SQL queries, reporting pipelines, and dashboards to support engineering and product decision-making using PowerBI, Tableau and Python.

Hardware Engineer, Satcom Direct — June 2019 – Jan 2023

- Designed and optimized multi-layer RF circuit boards for satellite-based in-flight connectivity systems, including simulation and RF parameter tuning using ADS and Altium Designer.
- Developed automated RF, Wi-Fi and wireless test workflows using Python and LabVIEW, evaluating signal integrity and system parameters across multi-functional engineering teams.

Publications

Handover Optimization in LEO Satellite Networks using Online Decision Transformer, IEEE Meditcom — July 2026

QoS Prediction for Satellite-Based Avionic Communication Using Transformers, IEEE TMLCN — Jan 2026